

LTA Natural-Tiling Ground-Gate Specification

Stage 10D: Exact Face Selection, Tile-Side Propagation, Convex Periodic Partition, and Bound Stability

mdstats

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Purpose and stage boundary

Stage 10D is the first end-to-end scientific ground gate for natural tilings. It certifies the natural tiling of the **unlabeled LTA framework net** at primitive ring bounds

$$K \in \{8, 10, 12\}.$$

The expected finite lifted tile types are

$$[4^6], \quad [4^{6.6^8}], \quad [4^{12.6^8.8^6}],$$

with primitive-cell multiplicities

$$6 : 2 : 2 = 3 : 1 : 1.$$

Runtime/API target:

mdstats 0.19.27a0

Primary module:

mdstats/analysis/lta_natural_tiling.py

The backend is intentionally **LTA-specific**. It is not promoted as a generic master-refinement constructor. It accepts only the exact unlabeled LTA quotient fingerprint and the complete validation sequence (8, 10, 12).

Scientific motivation

Stages 8C–10C supply the independent machinery required for natural tilings: embedded ring faces, translation-labelled cell complexes, partition evidence, properness, bounded face selection, and cross-bound comparison. None of those stages alone proves that the implementation recovers the published LTA natural tiling.

LTA is a strong end-to-end gate because increasing the ring bound from 8 to 12 introduces 32 additional primitive 12-rings. Those rings are boundedly strong in the active depth-one domain, so a gate that selects faces from strength alone would change the tiling incorrectly. The authoritative exact embedding provides the missing geometric distinction: all 32 new 12-rings are nonplanar and cannot be scientific 2-cells in the first planar-face backend.

External methods and original construction

The periodic quotient and translation-labelled cycle convention is compatible with the vector method of Chung, Hahn, and Klee [1]. The natural-tiling target and properness requirement follow Blatov, Delgado-Friedrichs, O’Keeffe, and Proserpio [2]. Exact sign decisions use rational arithmetic in the spirit of robust predicate design.

The following Stage-10D mechanisms are project-specific `mdstats` constructions:

- strict-convex-planar filtering of bounded-strong primitive rings in the authoritative rational embedding;
- exact cyclic sorting of incident face-interior rays in the quotient plane normal to a framework edge;
- translation-labelled face-side propagation into finite lifted tile shells;
- a convex-polytope periodic partition certificate based on exact supporting halfspaces, volumes, and separating axes;
- generator-based properness certification after exact replay of full-group closure; and
- proof-preserving reuse across larger K only after exact selected-ring stable keys are unchanged.

Input contract

Required topology

```
certify_lta_natural_tiling(  
    topology: FrameworkTopology,  
    *,  
    bounds: Sequence[int] = (8, 10, 12),  
    resources: LtaNaturalTilingResources | None = None,  
) -> LtaNaturalTilingGate
```

The first backend requires:

- one connected, rank-three, translation-index-one periodic net;
- 48 quotient vertices;
- 96 quotient edges;
- degree four at every quotient vertex;
- a complete exact unlabeled automorphism group of order 96; and
- the exact bound sequence (8, 10, 12).

A topology that merely has the expected ring counts is not accepted. Source identity, periodic rank, quotient size, degree sequence, and complete symmetry must all agree.

Finite resources

`LtaNaturalTilingResources` bounds:

- number of tested bounds;
- primitive rings per bound;
- selected scientific faces;
- exact fan self-intersection tests;

- exact framework–triangle contact tests;
- edge-sector incidences;
- periodic tile-image pairs;
- exact separating-axis tests;
- generator face and tile images; and
- generator-closure multiplication checks.

A resource failure raises `LtaNaturalTilingResourceError`; it is never converted into a scientific rejection.

Bound rebuild and stable reuse

For every K , Stage 10D independently rebuilds:

```
primitive rings
-> source-bound ring index
-> depth-one bounded strength
-> exact ring polygon geometry
-> selected scientific ring keys
```

At $K = 8$, the complete downstream stack is certified:

```
selected faces
-> embedded fan witnesses
-> exact edge-sector propagation
-> translation-labelled cell complex
-> full-group properness
-> convex periodic partition
```

For $K = 10$ and 12 , downstream evidence may be reused only after the new build proves byte-for-byte equality of the ordered selected `PrimitiveRingKey` digests. This is proof-preserving memoization, not stale source-object reuse. The higher bound has already independently rebuilt every stage that could introduce or remove a scientific face. The deterministic downstream construction depends only on:

- the fixed exact net view;
- the fixed authoritative embedding;
- the complete fixed automorphism group; and
- the exact selected ring-key set.

If those keys differ, the backend must perform a new downstream certification or reject the ground-gate assumption.

Bounded ring strength

For every primitive ring R of size n , the active domain is

```
RingStrengthDomain(
    target_ring_key=R.key,
    max_component_size=n - 1,
    placement_domain=EdgeIncidencePlacementDomain(1),
)
```

This tests decompositions into strictly smaller represented primitive rings whose placements lie within one exact edge-incidence step of the target.

Expected counts are:

Bound	Ring size	Strong in domain	Weak certified
8	4	36	0
8	6	16	24
8	8	6	0
10	same as 8	same	same
12	12	32	0

Strength is necessary but not sufficient for face selection.

Exact ring-polygon geometry

For a canonical lifted ring walk with exact fractional coordinates

$$\mathbf{p}_0, \dots, \mathbf{p}_{n-1},$$

Stage 10D first finds two independent in-plane vectors

$$\mathbf{u} = \mathbf{p}_i - \mathbf{p}_0, \quad \mathbf{v} = \mathbf{p}_j - \mathbf{p}_0.$$

Coplanarity requires

$$\det(\mathbf{u}, \mathbf{v}, \mathbf{p}_k - \mathbf{p}_0) = 0 \quad \text{for every } k.$$

An injective coordinate pair is then chosen from a nonzero 2×2 minor of $[\mathbf{u}\mathbf{v}]$. Every consecutive projected turn must be nonzero and have the same sign. The resulting states are:

STRICTLY_CONVEX_PLANAR
NONPLANAR
PLANAR_NONCONVEX_OR_DEGENERATE

A scientific face is selected exactly when

strength == STRONG_IN_DOMAIN
and geometry == STRICTLY_CONVEX_PLANAR

Expected selected faces at all three bounds are:

$$36 \times 4R, \quad 16 \times 6R, \quad 6 \times 8R.$$

At $K = 12$, the 32 new strong 12-rings are explicitly retained as strong-but-nonplanar exclusions.

Embedded face witnesses

Every selected strict convex planar n -gon uses the canonical fan

$$(0, 1, 2), (0, 2, 3), \dots, (0, n-2, n-1).$$

The backend verifies:

- nondegenerate exact triangles;
- no forbidden zero-image fan self-intersection;
- no forbidden periodic self-image intersection; and
- no framework-edge penetration other than the exact boundary edge or shared boundary vertex allowed by Stage 8C.

Unlike the general Stage-8C backend, Stage 10D does not enumerate every Catalan triangulation. Strict convex planarity proves that the canonical fan is an embedded disk in its supporting plane. Exact periodic and framework-contact tests supply the remaining global checks.

Exact face-side propagation

Incident face rays

For one lifted framework edge with direction $\mathbf{d}$ and midpoint $\mathbf{m}$, every incident selected face placement supplies a ray

$$\mathbf{r} = \mathbf{c}_F - \mathbf{m},$$

where $\mathbf{c}_F$ is the exact average of the lifted face vertices.

Choose a nonzero component d_k and the remaining axes a, b . A quotient-plane coordinate is

$$q(\mathbf{r}) = (d_k r_a - d_a r_k, d_k r_b - d_b r_k),$$

with one deterministic sign calibration so that the 2D cross-product order agrees with the 3D determinant

$$\det(\mathbf{d}, \mathbf{r}_i, \mathbf{r}_j).$$

Every LTA framework edge must have exactly three distinct incident selected-face rays.

Side adjacency and voltage gain

Consecutive cyclic rays define one local tile sector. For consecutive ring-edge occurrences A and B , Stage 10D joins the oriented face-side states

$$(F_A, s_A) \leftrightarrow (F_B, -s_B)$$

with translation gain

$$\mathbf{g} = \mathbf{t}_B - \mathbf{t}_A.$$

Translation offsets are propagated through each connected face-side component. A repeated node with a different accumulated offset proves a nonzero voltage cycle and rejects a slab, channel, or other noncompact region.

Each zero-gain component yields one finite lifted tile shell. If a component contains side state (F, s) at propagated translation $\mathbf{t}$, its boundary term is

$$-s[F, \mathbf{t}].$$

Translation-labelled cell complex

The 58 selected face orbits and ten propagated tile shells are passed to Stage 9. The exact complex must have

$$(N_0, N_1, N_2, N_3) = (48, 96, 58, 10),$$

and satisfy

$$\partial_1 \partial_2 = 0, \quad \partial_2 \partial_3 = 0, \quad 48 - 96 + 58 - 10 = 0.$$

Every face has exactly two translated tile-side incidences. Every lifted tile boundary is connected, nonbranching, orientable, and genus zero.

Properness from exact generators

The complete symmetry discovery stores a certified finite group G of order 96 and a reduced generator set S . Stage 10D independently replays closure of S under the stored multiplication table and requires

$$\langle S \rangle = G.$$

For every generator, the implementation maps every scientific face placement and every translation-labelled tile shell. If each generator preserves the complete scientific complex, every word in the generators preserves it; therefore every net automorphism preserves the tiling.

Because the tiling uses the fixed net as its 1-skeleton, this proves

$$\text{Aut}(\mathcal{T}) = \text{Aut}(G_{\text{view}}).$$

Auxiliary fan triangles and convex-partition support data do not participate in properness.

Exact convex periodic partition certificate

Convexity and interior points

For one tile shell, every oriented face supplies an outward plane

$$\mathbf{n}_F \cdot (\mathbf{x} - \mathbf{p}_F) \leq 0.$$

Every lifted tile vertex must satisfy every halfspace. The exact average of the distinct tile vertices must satisfy all inequalities strictly and is stored as an interior witness.

Exact volume

Each oriented polygon is triangulated by its fan. The signed fractional volume is

$$V_T = \frac{1}{6} \sum_{(\mathbf{a}, \mathbf{b}, \mathbf{c})} \mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}).$$

Every tile volume must be positive and

$$\sum_T V_T = 1.$$

For LTA, the exact per-tile volumes are:

Tile type	Multiplicity	Fractional volume
$[4^6]$	6	1/256
$[4^{6.6^8}]$	2	61/768
$[4^{12}.6^{8.8^6}]$	2	157/384

Periodic interior-disjointness

For every pair of tile orbits and every integer translation whose exact AABs can overlap, the certificate applies the convex-polytope separating-axis family:

- every face normal of either polytope; and
- every nonzero cross product of one edge direction from each polytope.

An axis $\mathbf{a}$ separates interiors when

$$\max_{x \in A} \mathbf{a} \cdot x \leq \min_{y \in B + \mathbf{t}} \mathbf{a} \cdot y$$

or the reversed inequality holds. Equality permits a shared boundary. If no axis separates a candidate pair, the construction is rejected as an interior overlap.

Pairwise periodic interior-disjointness plus exact total volume one excludes an open positive-measure void in the unit three-torus. This implication is used only after convexity and all periodic image candidates have been certified.

Persistent result model

```
LtaNaturalTilingGate(
    topology_graph_digest=...,
    periodic_net_view_digest=...,
    periodic_net_symmetry_digest=...,
    periodic_net_embedding_digest=...,
    bounds=(8, 10, 12),
    observations=...,
    expected_tile_multiplicities=...,
    selected_faces_stable=True,
```

```

    tiling_stable=True,
    expected_lta_match=True,
    status=LtaNaturalTilingGateStatus.CERTIFIED,
)

```

Each `LtaBoundObservation` records:

- primitive-ring counts by size;
- strength counts by size and state;
- exact geometry counts by size and state;
- selected face counts and stable ring-key digests;
- strong-but-nonplanar exclusions;
- cell counts;
- tile signatures and multiplicities;
- reduced multiplicity ratio;
- mesh-independent scientific complex key;
- exact convex partition volume, pair, and axis summaries; and
- full-group properness status.

`to_dict()` uses canonical JSON-compatible payloads and SHA-256 digests. `from_dict()` deterministically replays the complete gate against the supplied `FrameworkTopology` and rejects altered evidence.

Certification outcome

The gate is `CERTIFIED` only when all three bounds satisfy:

1. selected scientific ring keys are identical;
2. the mesh-independent scientific tile complex is identical;
3. cell counts are (48, 96, 58, 10);
4. tile multiplicities are 6, 2, 2 in the expected signature order;
5. the reduced ratio is 3 : 1 : 1;
6. exact convex periodic partition certification succeeds; and
7. the complete unlabeled net automorphism group preserves the tiling.

Any proved mismatch is `REJECTED`. A future generalized backend may return `UNRESOLVED` for bounded or unsupported cases; the present strict ground gate raises explicit input/resource errors before constructing a misleading result.

Edge cases and failure policy

The implementation rejects:

- any bound sequence other than (8, 10, 12);
- a non-LTA quotient fingerprint;
- incomplete or non-order-96 symmetry;
- an uncertified straight-edge embedding;
- selected faces with nonplanarity, degeneracy, self-intersection, or framework penetration;
- an edge with other than three distinct selected face sectors;
- nonzero translation cycles in a putative tile component;
- a cell complex that violates chain, incidence, Euler, or shell invariants;
- a generator image outside the selected face or tile set;
- nonconvex tile shells;
- nonpositive or nonclosing exact tile volumes;
- periodic tile interior overlap; and
- altered persistent digests or replay payloads.

Focused validation requirements

The Stage-10D focused gate must verify:

- complete certification at $K = 8, 10, 12$;
- ring counts 82, 82, 114;
- 32 new nonplanar 12-rings at $K = 12$;
- stable selected faces $36 \times 4R$, $16 \times 6R$, $6 \times 8R$;
- tile counts 6, 2, 2 and ratio 3 : 1 : 1;
- exact cell counts (48, 96, 58, 10);
- exact tile volumes and total volume one;
- periodic SAT candidate and axis audit counts;
- generator closure and scientific-complex preservation;
- resource preflight;
- exact bound-sequence rejection; and
- digest tamper rejection.

Limitations and next boundary

Stage 10D proves the expected LTA result under one exact, strict-convex-planar first backend. It does not yet:

- construct a generic master refinement from arbitrary strong rings;
- resolve nonplanar or multiply embedded face surfaces;
- search alternative witness families;
- recognize arbitrary nets as LTA up to unrestricted periodic isomorphism;
- prove stabilization for every $K > 12$; or
- convert tiles automatically into physical cages, portals, and accessibility networks.

The next implementation boundary is Stage 11: tile geometry, cage identity, windows/portals, and chemistry-aware accessibility built on the now-certified LTA natural tiling.

References

- [1] S. J. Chung, T. Hahn, and W. E. Klee, “Nomenclature and generation of three-periodic nets: the vector method”, *Acta Crystallographica A* **40**, 42–50 (1984). doi:10.1107/S010876738400010X.
- [2] V. A. Blatov, O. Delgado-Friedrichs, M. O’Keeffe, and D. M. Proserpio, “Three-periodic nets and tilings: natural tilings for nets”, *Acta Crystallographica A* **63**, 418–425 (2007). doi:10.1107/S0108767307038287.